

ADI KARTHIKEYA SUNKARA BALASUBRAMANYAM

Houston, TX | (832) 654-1709 | adikarthikeya1234@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

- AI/ML Engineer (M.S. Data Science, May 2026) with hands-on experience building agentic AI systems, RAG pipelines, and applied machine learning models across healthcare, e-commerce, energy, and consumer domains.
- Skilled in LLM orchestration (LangGraph, Google ADK), classical ML (XGBoost, Random Forest, SHAP), and full-stack deployment (FastAPI, Streamlit, Docker).
- Proven ability to translate technical results into stakeholder-ready insight, from a federally funded NSF research bootcamp to award-ranked hackathon builds and university-level ML instruction.

SKILLS

AI & ML: LLMs (Claude, Gemini, Groq), Agentic Workflows (LangGraph, Google ADK), RAG & Retrieval Systems, Scikit-learn, XGBoost, LightGBM, TensorFlow, SHAP, NLP (Transformers, VADER)

Languages & Data: Python, SQL, R, Pandas, NumPy

Deployment & Tools: Docker, FastAPI, Streamlit, Git, Vercel, Render

Data Engineering & DBs: PostgreSQL, ChromaDB, Neo4j, Tableau, QGIS, Orange ML

EDUCATION

University of Houston

M.S. Data Science, GPA 3.9/4.0

Houston, TX

May 2026

Amrita Vishwa Vidyapeetham

B.Tech, Electronics and Communication Engineering, GPA 3.50/4.00

Bengaluru, India

May 2024

EXPERIENCE

Teaching Assistant

Jan 2025 – May 2026

University of Houston

Houston, TX

- Supported 120+ students across Pattern Recognition and Machine Learning (CEA 4341), Engineering Data Analytics (CEA 4355), and data analytics/business courses through office hours, grading, and review sessions.
- Reworked assignment rubrics for CEA 4341, cutting grading turnaround by roughly 30% across the cohort.
- Led discussion and debugging sessions, guiding students through end-to-end classification projects in Python and scikit-learn – EDA, model selection, and evaluation.

Data Science Intern, NSF Data Science for Energy Transition (Federally Funded)

May 2025 – Jun 2025

University of Houston

Remote

- Selected for a nationally funded 5-week NSF bootcamp in data science for energy, environment, and geoscience, working alongside researchers and industry professionals on real energy datasets.
- Built facies classification models (KNN, Random Forest, Decision Tree), scoring 0.8024 accuracy in a national Kaggle-style competition.
- Applied PCA and K-Means on Tellus geophysics data to interpret subsurface terranes, and trained a TensorFlow model to predict rock fertility from 19-element zircon geochemistry.
- Built interactive Tableau dashboards (EIA energy production, U.S. Census data) and QGIS maps of geothermal/solar potential for research stakeholders.

PROJECTS

RareDx, Agentic Rare Disease Diagnostic Assistant | LangGraph, Neo4j, ChromaDB, Groq

Mar 2026 – May 2026

- Engineered an AI diagnostic workflow in LangGraph, enriching ChromaDB with full Orphanet disease profiles using BioLORD embeddings, lifting Top-1 accuracy from 10% to 92.5% and Recall@5 to 100% across 40 adversarial cases.
- Designed ablation experiments across 7 retrieval strategies (280 evaluations) and a confidence calibration pipeline (Brier 0.068, ECE 0.051), achieving zero hallucination across all outputs.

ShopperAI, Agentic AI Shopping Assistant | Python, Google ADK, Gemini, LangChain

Dec 2025

- Built a deterministic, stage-based agentic shopping assistant guiding users through product discovery and virtual try-on, ranking 4th globally out of 200 teams at the ODSC AI & Google Cloud Agentic AI Hackathon.

Question Answering System (NLP) | BERT, RoBERTa, DeBERTa, Flan-T5

Sep 2025 – Dec 2025

- Fine-tuned four transformer architectures on SQuAD v2, improving Exact Match from 34.9% to 75.3% and reaching F1 scores up to 78.4%; built a real-time interface to compare model predictions.

CropGuard AI, Crop Disease Diagnosis | Next.js, Groq (LLM), TF-IDF, Open-Meteo

2026

- Built a free, symptom-based diagnostic tool grounded via TF-IDF retrieval over a curated 31-disease knowledge base, eliminating hallucinated advice, backed by live weather and climate-history data.

Predictive Match ML Dashboard | XGBoost, LightGBM, SHAP, Streamlit, NLP

Jun 2026

- Engineered 52 features across 60,000 dating profiles, stacked ensemble models, and shipped a Streamlit Profile Optimizer using SHAP explainability to rank personalized improvements.